Example 1

Assumptions

1. Have some training data $\mathcal{D} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\}$
2. Assume a prior distribution $P(\boldsymbol{\theta})$
   $P(\boldsymbol{\theta}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$
3. Design a likelihood function $P(y|\mathbf{x}, \boldsymbol{\theta})$
   $P(y|\mathbf{x}, \boldsymbol{\theta}) = f_{NN}(\mathbf{x}; \boldsymbol{\theta})|_y$ --a softmax neural network
4. Decide on approximating distribution $Q_{\boldsymbol{\omega}}(\boldsymbol{\theta})$
   $Q_{\boldsymbol{\omega}}(\boldsymbol{\theta}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2\mathbf{I})$
5. Find $\boldsymbol{\omega}$ that optimizes the following objective

We cannot get rid of the expectation anymore!

$$\max_{\boldsymbol{\omega}} \sum \left(Q_{\boldsymbol{\omega}}(\boldsymbol{\theta}) \sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}) \right) - D_{KL}\left(Q_{\boldsymbol{\omega}}(\boldsymbol{\theta}) \,\|\, P(\boldsymbol{\theta})\right)$$



Example 1

Approximating Expectations

- Different sampling techniques are approximations to expensive or intractable expectations, $\mathbb{E}$.

- Simplest sampling-based approximation is the following Monte Carlo integration:

$$\mathbb{E}_{Q(\mathbf{x})}[f(\mathbf{x})] = \sum_{\mathbf{x}} Q(\mathbf{x}) f(\mathbf{x}) \simeq \frac{1}{s} \sum_{k=1}^{s} f(\mathbf{x}_k)$$

$$\{\mathbf{x}_k\} \sim Q(\mathbf{x})$$



Approximating Expectations

1. Have some training data $\mathcal{D} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\}$
2. Assume a prior distribution $P(\boldsymbol{\theta})$
   $P(\boldsymbol{\theta}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$
3. Design a likelihood function $P(y|\mathbf{x}, \boldsymbol{\theta})$
   $P(y|\mathbf{x}, \boldsymbol{\theta}) = f_{NN}(\mathbf{x}; \boldsymbol{\theta})|_y$ --a softmax neural network
4. Decide on approximating distribution $Q_{\boldsymbol{\omega}}(\boldsymbol{\theta})$
   $Q_{\boldsymbol{\omega}}(\boldsymbol{\theta}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2\mathbf{I})$
5. Find $\boldsymbol{\omega}$ that optimizes the following objective

Let's look at this now

$$\max_{\boldsymbol{\omega}} \frac{1}{s}\sum_{k=1}^{s}\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k) - D_{KL}(Q_{\boldsymbol{\omega}}(\boldsymbol{\theta}) \,\|\, P(\boldsymbol{\theta}))$$

sampling $\quad \{\boldsymbol{\theta}_k\}_{k=1:s} \sim Q_{\boldsymbol{\omega}}(\boldsymbol{\theta})$

$$\max_{\boldsymbol{\omega}} \sum_{\boldsymbol{\theta}} \left(Q_{\boldsymbol{\omega}}(\boldsymbol{\theta}) \sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}) \right) - D_{KL}(Q_{\boldsymbol{\omega}}(\boldsymbol{\theta}) \,\|\, P(\boldsymbol{\theta}))$$

KL-divergence of two Gaussian distributions

Interlude: KL-divergence of two Gaussian distributions

$$D_{KL}\left(\mathcal{N}(\mu_1, \sigma_1^2) \parallel \mathcal{N}(\mu_2, \sigma_2^2)\right)$$

$$= \sum \mathcal{N}(\mu_1, \sigma_1^2) \log \frac{\mathcal{N}(\mu_1, \sigma_1^2)}{\mathcal{N}(\mu_2, \sigma_2^2)}$$

$$= \sum \mathcal{N}(\mu_1, \sigma_1^2) \log \mathcal{N}(\mu_1, \sigma_1^2) - \sum \mathcal{N}(\mu_1, \sigma_1^2) \log \mathcal{N}(\mu_2, \sigma_2^2)$$

$$= -\frac{1}{2}(1 + log\, 2\pi\sigma_1^2) + \frac{1}{2}(log\, 2\pi\sigma_2^2) + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2}$$

$$= \log \frac{\sigma_2}{\sigma_1} - \frac{1}{2} + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2}$$

We have closed-form solution for KL-divergence of two Gaussian distributions



Example 1

Inserting KL-divergence of two Gaussian distributions

1. Have some training data $\mathcal{D} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\}$
2. Assume a prior distribution $P(\boldsymbol{\theta})$
   $P(\boldsymbol{\theta}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$
3. Design a likelihood function $P(y|\mathbf{x}, \boldsymbol{\theta})$
   $P(y|\mathbf{x}, \boldsymbol{\theta}) = f_{NN}(\mathbf{x}; \boldsymbol{\theta})|_y$ --a softmax neural network
4. Decide on approximating distribution $Q_{\boldsymbol{\omega}}(\boldsymbol{\theta})$
   $Q_{\boldsymbol{\omega}}(\boldsymbol{\theta}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2\mathbf{I})$
5. Find $\boldsymbol{\omega}$ that optimizes the following objective

$$\boldsymbol{\sigma} = (\sigma_1, \sigma_2, \ldots, \sigma_p)$$

$$\max_{\boldsymbol{\omega}} \frac{1}{S}\sum_{k=1}^{S}\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k) + \sum_{j=1}^{p}\left(\log \sigma_j - \frac{\sigma_j^2 + \mu_j^2}{2}\right)$$

D_{KL} of Gaussians ↑ diagonal covariance matrix

$$\max_{\boldsymbol{\omega}} \frac{1}{S}\sum_{k=1}^{S}\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k) - D_{KL}(Q_{\boldsymbol{\omega}}(\boldsymbol{\theta}) \,\|\, P(\boldsymbol{\theta}))$$

sampling ↑ $\{\boldsymbol{\theta}_k\} \sim Q_{\boldsymbol{\omega}}(\boldsymbol{\theta})$

$$\max_{\boldsymbol{\omega}} \sum\left(Q_{\boldsymbol{\omega}}(\boldsymbol{\theta})\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta})\right) - D_{KL}(Q_{\boldsymbol{\omega}}(\boldsymbol{\theta}) \,\|\, P(\boldsymbol{\theta}))$$

Now, we can calculate the objective, but we need the derivates too!

Example 1

Derivatives Our objective: $\min_{\boldsymbol{\omega}} \mathcal{L}(\boldsymbol{\omega}) \triangleq -\frac{1}{s}\sum_{\boldsymbol{k=1}}^{\boldsymbol{s}}\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k) - \sum_{j=1}^{p}\left(\log \sigma_j - \frac{\sigma_j^2 + \mu_j^2}{2}\right)$

with $\boldsymbol{\theta}_k \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma^2 \mathrm{I}})$

For standard back-propagation, we would need the first derivative:

$\frac{\partial \mathcal{L}(\boldsymbol{\omega}=\{\boldsymbol{\mu},\boldsymbol{\sigma^2}\})}{\partial \boldsymbol{\mu}}$ and $\frac{\partial \mathcal{L}(\boldsymbol{\omega}=\{\boldsymbol{\mu},\boldsymbol{\sigma^2}\})}{\partial \boldsymbol{\sigma^2}}$

How can we get the derivatives of monte-carlo integration w.r.t. distribution parameters?

Derivatives of Gaussian's Expectation

Derivatives

- For derivatives of expectation of multivariate Gaussian distributions we have:

$$\frac{\partial \mathbb{E}_{\mathbf{x}\sim\mathcal{N}}[g(\mathbf{x})]}{\partial \boldsymbol{\mu}} = \mathbb{E}_{\mathbf{x}\sim\mathcal{N}}\left[\frac{\partial g(\mathbf{x})}{\partial \mathbf{x}}\right]$$

$$\frac{\partial \mathbb{E}_{\mathbf{x}\sim\mathcal{N}}[g(\mathbf{x})]}{\partial \boldsymbol{\sigma}^2} = \frac{1}{2}\mathbb{E}_{\mathbf{x}\sim\mathcal{N}}\left[\frac{\partial^2 g(\mathbf{x})}{\partial \mathbf{x}^2}\right]$$

sampling ↓ $\{\mathbf{x}_k\} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2)$

$$\frac{\partial \mathbb{E}_{\mathbf{x}\sim\mathcal{N}}[g(\mathbf{x})]}{\partial \boldsymbol{\mu}} \simeq \frac{1}{s}\sum_{k=1}^{s}\frac{\partial g(\mathbf{x}_k)}{\partial \mathbf{x}}$$

$$\frac{\partial \mathbb{E}_{\mathbf{x}\sim\mathcal{N}}[g(\mathbf{x})]}{\partial \boldsymbol{\sigma}^2} \simeq \frac{1}{2s}\sum_{k=1}^{s}\left(\frac{\partial^2 g(\mathbf{x}_k)}{\partial \mathbf{x}^2}\right)$$

[Opper and Archambeau. "The variational gaussian approximation revisited". Neural Computation 2009]

Derivatives

Our objective: $\min_{\boldsymbol{\omega}} \mathcal{L}(\boldsymbol{\omega}) \triangleq -\frac{1}{s}\sum_{k=1}^{s}\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k) - \sum_{j=1}^{p}\left(\log \sigma_j - \frac{\sigma_j^2 + \mu_j^2}{2}\right)$

with $\boldsymbol{\theta}_k \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2\mathbf{I})$

For backpropagation, we would need the first derivative:

$$\frac{\partial \mathcal{L}(\boldsymbol{\omega} = \{\boldsymbol{\mu}, \boldsymbol{\sigma}^2\})}{\partial \mu_i} = \frac{\partial\left(-\mathbb{E}_{\boldsymbol{\theta}\sim\mathcal{N}}[\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta})] - \sum_{j=1}^{p}\left(\log \sigma_j + \frac{\sigma_j^2 + \mu_j^2}{2}\right)\right)}{\partial \mu_i}$$

$$= \frac{\partial(-\mathbb{E}_{\boldsymbol{\theta}\sim\mathcal{N}}[\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta})])}{\partial \mu_i} - \frac{\partial\left(\sum_{j=1}^{p}\left(\log \sigma_j + \frac{\sigma_j^2 + \mu_j^2}{2}\right)\right)}{\partial \mu_i}$$

$$= \frac{\partial(-\mathbb{E}_{\boldsymbol{\theta}\sim\mathcal{N}}[\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta})])}{\partial \mu_i} + \frac{\partial\left(\frac{\mu_i^2}{2}\right)}{\partial \mu_i}$$

$$= \frac{\partial(-\mathbb{E}_{\boldsymbol{\theta}\sim\mathcal{N}}[\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta})])}{\partial \mu_i} + \mu_i$$

Derivatives

Our objective: $\min_{\boldsymbol{\omega}} \mathcal{L}(\boldsymbol{\omega}) \triangleq -\frac{1}{s}\sum_{k=1}^{s}\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k) - \sum_{j=1}^{p}\left(\log \sigma_j + \frac{\sigma_j^2 + \mu_j^2}{2}\right)$

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$$= -\frac{1}{s}\sum_{k=1}^{s}\left(\frac{\partial(\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k))}{\partial \boldsymbol{\theta}_i}\right) + \mu_i$$

$$= -\frac{1}{s}\sum_{k=1}^{s}\sum_{i=1}^{n}\left(\frac{\partial\left(\log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k)\right)}{\partial \boldsymbol{\theta}_i}\right) + \mu_i$$

This is similar to the gradient computed for cross entropy!

Derivatives

Our objective: $\min_{\boldsymbol{\omega}} \mathcal{L}(\boldsymbol{\omega}) \triangleq -\frac{1}{s}\sum_{k=1}^{s}\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k) - \sum_{j=1}^{p}\left(\log \sigma_j - \frac{\sigma_j^2 + \mu_j^2}{2}\right)$

with $\boldsymbol{\theta}_k \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2\mathbf{I})$

For backpropagation, we would need the first derivative:

$$\frac{\partial \mathcal{L}(\boldsymbol{\omega} = \{\boldsymbol{\mu}, \boldsymbol{\sigma}^2\})}{\partial \sigma_i^2} = \frac{\partial\left(-\mathbb{E}_{\boldsymbol{\theta}\sim\mathcal{N}}[\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta})] - \sum_{j=1}^{p}\left(\log \sigma_j - \frac{\sigma_j^2 + \mu_j^2}{2}\right)\right)}{\partial \sigma_i^2}$$

$$= \frac{\partial(-\mathbb{E}_{\boldsymbol{\theta}\sim\mathcal{N}}[\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta})])}{\partial \sigma_i^2} - \frac{\partial\left(\sum_{j=1}^{p}\left(\log \sigma_j - \frac{\sigma_j^2 + \mu_j^2}{2}\right)\right)}{\partial \sigma_i^2}$$

$$= \frac{\partial(-\mathbb{E}_{\boldsymbol{\theta}\sim\mathcal{N}}[\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta})])}{\partial \sigma_i^2} - \frac{\partial\left(\log \sigma_i - \frac{\sigma_i^2}{2}\right)}{\partial \sigma_i^2}$$

$$= \frac{\partial(-\mathbb{E}_{\boldsymbol{\theta}\sim\mathcal{N}}[\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta})])}{\partial \sigma_i^2} - \frac{\partial(\log \sigma_i)}{\partial \sigma_i^2} + \frac{\partial\left(\frac{\sigma_i^2}{2}\right)}{\partial \sigma_i^2}$$

$$= \frac{\partial(-\mathbb{E}_{\boldsymbol{\theta}\sim\mathcal{N}}[\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta})])}{\partial \sigma_i^2} - \frac{1}{2\sigma_i^2} + \frac{1}{2}$$

Derivatives

Our objective: $\min_{\boldsymbol{\omega}} \mathcal{L}(\boldsymbol{\omega}) \triangleq -\frac{1}{s}\sum_{k=1}^{s}\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k) - \sum_{j=1}^{p}\left(\log \sigma_j - \frac{\sigma_j^2 + \mu_j^2}{2}\right)$

with $\boldsymbol{\theta}_k \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2\mathbf{I})$

For backpropagation, we would need the first derivative:

$$\frac{\partial \mathcal{L}(\boldsymbol{\omega} = \{\boldsymbol{\mu}, \boldsymbol{\sigma}^2\})}{\partial \sigma_i^2} = \frac{\partial(-\mathbb{E}_{\boldsymbol{\theta}\sim\mathcal{N}}[\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta})])}{\partial \sigma_i^2} - \frac{1}{2\sigma_i^2}$$

$$= -\frac{1}{2s}\sum_{k=1}^{s}\left(\frac{\partial^2(\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k))}{\partial \boldsymbol{\theta}_i^2}\right) - \frac{1}{2\sigma_i^2}$$

$$= -\frac{1}{2s}\sum_{k=1}^{s}\sum_{i=1}^{n}\frac{\partial^2(\log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k))}{\partial \boldsymbol{\theta}_i^2} - \frac{1}{2\sigma_i^2}$$

Optimization

We would like to find the best values variational parameters $\boldsymbol{\omega}^* = \{\boldsymbol{\mu}^*, \boldsymbol{\sigma^2}^*\}$

We have the VA objective

$$\min_{\boldsymbol{\omega}} \mathcal{L}(\boldsymbol{\omega}) \triangleq -\frac{1}{s}\sum_{k=1}^{s}\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k) - \sum_{j=1}^{p}\left(\log \sigma_j - \frac{\sigma_j^2 + \mu_j^2}{2}\right)$$

$$\text{with } \boldsymbol{\theta}_k \sim \mathcal{N}\left(\boldsymbol{\mu}, \boldsymbol{\sigma^2}\mathbf{I}\right)$$

We have the derivatives of the objective w.r.t. variational parameters

$$\frac{\partial \mathcal{L}(\boldsymbol{\omega} = \{\boldsymbol{\mu}, \boldsymbol{\sigma^2}\})}{\partial \mu_i} = -\frac{1}{s}\sum_{k=1}^{s}\sum_{i=1}^{n}\left(\frac{\partial\left(\log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k)\right)}{\partial \boldsymbol{\theta}_i}\right) + \mu_i$$

$$\frac{\partial \mathcal{L}(\boldsymbol{\omega} = \{\boldsymbol{\mu}, \boldsymbol{\sigma^2}\})}{\partial \sigma_i^2} = -\frac{1}{2s}\sum_{k=1}^{s}\sum_{i=1}^{n}\frac{\partial^2\left(\log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k)\right)}{\partial \boldsymbol{\theta}_i^2} - \frac{1}{2\sigma_i^2}$$

Example 1

Optimization

- In deep networks, we usually have
  - $n \in O(10^4)$ or $O(10^6)$
  - $p \in O(10^4)$ or $O(10^6)$
- So, it will be more feasible with stochastic gradient descent

$$\min_{\boldsymbol{\omega}} \mathcal{L}(\boldsymbol{\omega}) \triangleq -\frac{1}{s}\sum_{k=1}^{s}\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k) - \sum_{j=1}^{p}\left(\log \sigma_j - \frac{\sigma_j^2 + \mu_j^2}{2}\right)$$

with $\boldsymbol{\theta}_k \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2\mathbf{I})$

$$\frac{\partial \mathcal{L}(\boldsymbol{\omega} = \{\boldsymbol{\mu}, \boldsymbol{\sigma}^2\})}{\partial \mu_i} = -\frac{1}{s}\sum_{k=1}^{s}\sum_{i=1}^{n}\left(\frac{\partial(\log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k))}{\partial \boldsymbol{\theta}_i}\right) + \mu_i$$

$$\frac{\partial \mathcal{L}(\boldsymbol{\omega} = \{\boldsymbol{\mu}, \boldsymbol{\sigma}^2\})}{\partial \sigma_i^2} = -\frac{1}{2s}\sum_{k=1}^{s}\sum_{i=1}^{n}\frac{\partial^2(\log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k))}{\partial \boldsymbol{\theta}_i^2} - \frac{1}{2\sigma_i^2}$$



$$\frac{\partial \mathcal{L}(\boldsymbol{\omega} = \{\boldsymbol{\mu}, \boldsymbol{\sigma}^2\})}{\partial \mu_i} = -\frac{n}{sn_b}\sum_{k=1}^{s}\sum_{i=1}^{n_b}\left(\frac{\partial(\log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k))}{\partial \boldsymbol{\theta}_i}\right) + \mu_i$$

$$\frac{\partial \mathcal{L}(\boldsymbol{\omega} = \{\boldsymbol{\mu}, \boldsymbol{\sigma}^2\})}{\partial \sigma_i^2} = -\frac{n}{2sn_b}\sum_{k=1}^{s}\sum_{i=1}^{n_b}\frac{\partial^2(\log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k))}{\partial \boldsymbol{\theta}_i^2} - \frac{1}{2\sigma_i^2}$$

Example 1

Optimization

- Second-order derivatives are expensive, let's make another approximation

$$\min_{\boldsymbol{\omega}} \mathcal{L}(\boldsymbol{\omega}) \triangleq -\frac{1}{s}\sum_{k=1}^{s}\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k) - \sum_{j=1}^{p}\left(\log \sigma_j - \frac{\sigma_j^2 + \mu_j^2}{2}\right)$$

with $\boldsymbol{\theta}_k \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2\mathbf{I})$

$$\frac{\partial \mathcal{L}(\boldsymbol{\omega} = \{\boldsymbol{\mu}, \boldsymbol{\sigma}^2\})}{\partial \mu_i} = -\frac{n}{s n_b}\sum_{k=1}^{s}\sum_{i=1}^{n_b}\left(\frac{\partial\left(\log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k)\right)}{\partial \boldsymbol{\theta}_i}\right) + \mu_i$$

$$\frac{\partial \mathcal{L}(\boldsymbol{\omega} = \{\boldsymbol{\mu}, \boldsymbol{\sigma}^2\})}{\partial \sigma_i^2} = -\frac{n}{2 s n_b}\sum_{k=1}^{s}\sum_{i=1}^{n_b}\frac{\partial^2\left(\log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k)\right)}{\partial \boldsymbol{\theta}_i^2} - \frac{1}{2\sigma_i^2}$$

$$\frac{\partial \mathcal{L}(\boldsymbol{\omega} = \{\boldsymbol{\mu}, \boldsymbol{\sigma}^2\})}{\partial \sigma_i^2} \cong -\frac{n}{2 s n_b}\sum_{k=1}^{s}\sum_{i=1}^{n_b}\left(\frac{\partial\left(\log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k)\right)}{\partial \boldsymbol{\theta}_i}\right)^2 - \frac{1}{2\sigma_i^2}$$

Example 1

Optimization

- Each sample $\boldsymbol{\theta}_k$ of $\mathcal{N}\left(\boldsymbol{\mu}, \boldsymbol{\sigma}^2\mathbf{I}\right)$ requires a forward pass for the whole batch

- Let's approximate it further by using the same parameter sample $\boldsymbol{\theta}_k$ for each sample in the batch

$$\min_{\boldsymbol{\omega}} \mathcal{L}(\boldsymbol{\omega}) \triangleq -\frac{1}{s}\sum_{k=1}^{s}\sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k) - \sum_{j=1}^{p}\left(\log \sigma_j - \frac{\sigma_j^2 + \mu_j^2}{2}\right)$$

with $\boldsymbol{\theta}_k \sim \mathcal{N}\left(\boldsymbol{\mu}, \boldsymbol{\sigma}^2\mathbf{I}\right)$

$$\frac{\partial \mathcal{L}\left(\boldsymbol{\omega} = \{\boldsymbol{\mu}, \boldsymbol{\sigma}^2\}\right)}{\partial \mu_i} = -\frac{n}{s n_b}\sum_{k=1}^{s}\sum_{i=1}^{n_b}\left(\frac{\partial\left(\log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k)\right)}{\partial \boldsymbol{\theta}_i}\right) + \mu_i$$

$$\frac{\partial \mathcal{L}\left(\boldsymbol{\omega} = \{\boldsymbol{\mu}, \boldsymbol{\sigma}^2\}\right)}{\partial \sigma_i^2} \cong -\frac{n}{2 s n_b}\sum_{k=1}^{s}\sum_{i=1}^{n_b}\left(\frac{\partial\left(\log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k)\right)}{\partial \boldsymbol{\theta}_i}\right)^2 - \frac{1}{2\sigma_i^2}$$



$$\frac{\partial \mathcal{L}\left(\boldsymbol{\omega} = \{\boldsymbol{\mu}, \boldsymbol{\sigma}^2\}\right)}{\partial \mu_i} = -\frac{n}{n_b}\sum_{i=1}^{n_b}\left(\frac{\partial\left(\log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k)\right)}{\partial \boldsymbol{\theta}_i}\right) + \mu_i$$

$$\frac{\partial \mathcal{L}\left(\boldsymbol{\omega} = \{\boldsymbol{\mu}, \boldsymbol{\sigma}^2\}\right)}{\partial \sigma_i^2} \cong -\frac{n}{2 n_b}\sum_{i=1}^{n_b}\left(\frac{\partial\left(\log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}_k)\right)}{\partial \boldsymbol{\theta}_i}\right)^2 - \frac{1}{2\sigma_i^2}$$

Example 1

Optimization algorithm

- Initialize all variational parameters $\boldsymbol{\omega^0} = \{\boldsymbol{\mu^0}, \boldsymbol{\sigma^{2^0}}\}$

- For t=1:T iterations
  - sample from the variational distribution $\boldsymbol{\theta}_k \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma^2 \mathrm{I}})$
  - sample a minibatch $B = \{(\mathbf{x}_i, y_i)\}_{i=1:n_b}$
  - make a forward pass all the batch samples $f_{NN}(\mathbf{x}_{i=1:n_b}; \boldsymbol{\theta})$
  - calculate the loss for the batch samples $\mathcal{L}(\boldsymbol{\omega})$ with $P(y|\mathbf{x}, \boldsymbol{\theta}) = f_{NN}(\mathbf{x}; \boldsymbol{\theta})|_y$
  - backward pass and update variational parameters

$$\boldsymbol{\omega^t} = \{\boldsymbol{\mu^{t-1}} - \boldsymbol{\lambda}\frac{\partial \mathcal{L}}{\partial \boldsymbol{\mu}}|_{\boldsymbol{\omega}=\{\boldsymbol{\mu}^{t-1}, \boldsymbol{\sigma}^{2^{t-1}}\}}, \boldsymbol{\sigma^{2^{t-1}}} - \boldsymbol{\lambda}\frac{\partial \mathcal{L}}{\partial \boldsymbol{\sigma^2}}|_{\boldsymbol{\omega}=\{\boldsymbol{\mu}^{t-1}, \boldsymbol{\sigma}^{2^{t-1}}\}}\}$$



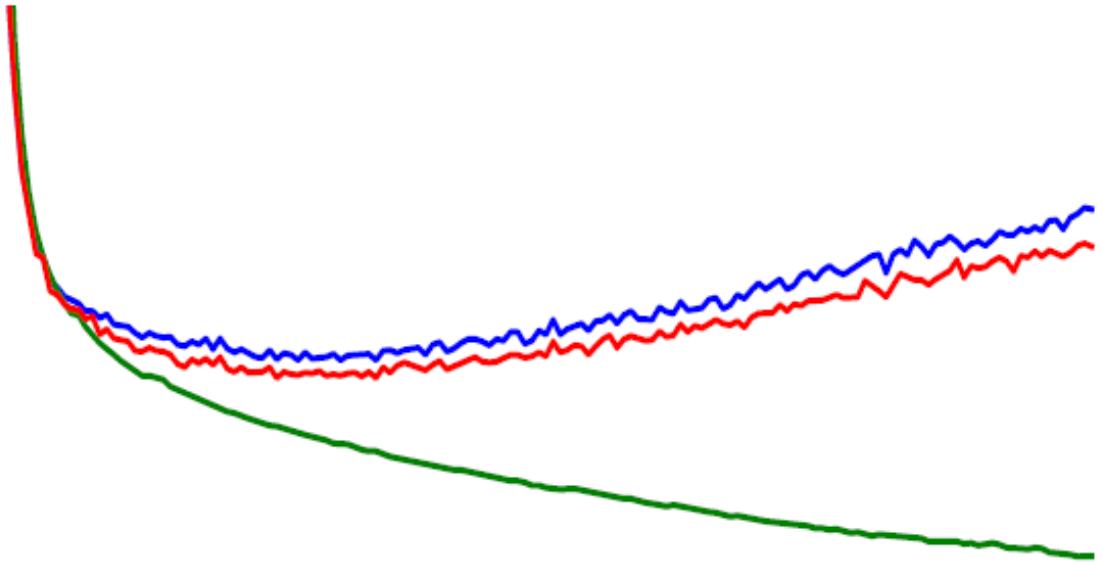
Example 1

[Alex Graves “Practical Variational Inference for Neural Networks”, NIPS 2011]



Example 1

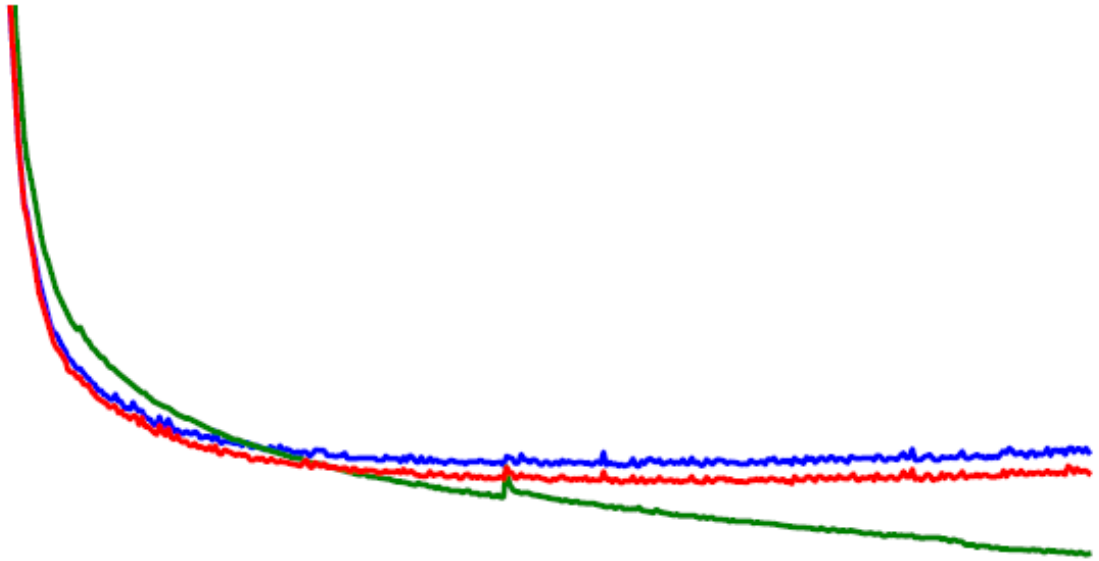
Results



Maximum likelihood

Example 1

Results



Assuming has fixed variance, the training becomes similar to adding noise during training!

Example 1

Some points of improvements

- We used variational approximation
- We assumed parameters are Gaussian-distributed.
- We assumed the parameters are independent (*i.e.*, approx. posterior's Gaussian distribution has diagonal covariance matrix)
- We used Monte Carlo integration to approximate the expectation in VA's expected likelihood
- We approximated the 2nd order derivatives
- The optimization is costly (requires sampling – multiple forward and backward passes)

